

# Rating Formula 1

Joint order statistics, a disaster slab,  
and seventy years of mechanical attrition

Peter Cotton

July 9, 2026

## Abstract

Formula 1 is a hostile environment for rating systems: grids of ten to fifty-five cars, full finish orders, and a retirement process that historically eliminated half the field. We rate every world-championship grand prix from 1950 to the present (1,158 races) under a prequential protocol — predict, score, then update — using a lattice-based Thurstonian system whose beliefs are unrestricted densities and whose update uses the exact likelihood of the whole finish order, computed in  $O(N)$  per contestant. It beats TrueSkill, Elo, Glicko-2 and OpenSkill on log loss, accuracy and rank correlation, while the pairwise-decomposition systems degrade or diverge as grids widen: Glicko-2 scores below the uniform baseline. Because the lattice imposes no distributional family, the performance density is a modeling choice, and the right choice is not Gaussian: adding a *disaster slab* — a lump of far-slow mass matching the retirement process — improves every metric, and letting the slab mass track the trailing retirement rate sets the overall record while its fitted value follows the sport’s reliability revolution from 59% attrition in the 1980s to 13% today. We report what failed candidly: symmetric heavy tails hurt (they invent impossibly fast miracles), skew does nothing anywhere, and a near-deterministic pace core is refuted — pace noise among finishers is genuinely order one. What survives is a taxonomy: model catastrophe frequency, not time-domain shape.

## 1 Why Formula 1

Most rating systems were built for head-to-head play and extended to multi-entrant events by decomposition: a finish order becomes a bag of pairwise results (Elo, Glicko-2, the Weng–Lin variants) or a chain of adjacent factors (TrueSkill). At tennis scale the decomposition is harmless. At Formula 1 scale it is the whole question. A grid of twenty cars implies 190 pairwise results whose errors are far from independent, and the sport’s history includes grids of fifty-five.

Formula 1 also embeds a distributional regime change inside a single dataset. The fraction of entrants failing to finish, by decade:

| decade   | 1950s | 60s | 70s | 80s | 90s | 2000s | 10s | 20s |
|----------|-------|-----|-----|-----|-----|-------|-----|-----|
| DNF rate | .42   | .49 | .51 | .59 | .51 | .30   | .18 | .13 |

No fixed distributional assumption is right for both halves of this table. A rating system whose noise model is a free parameter can say so; one locked to a Gaussian cannot.

The data (the community-maintained fldb compilation, CC-BY) is clean about edge cases that matter: drivers stopping past 90% distance are classified with positions; earlier retirements carry no position and enter our protocol tied at the back of the field; shared drives in the 1950s produce genuine dead heats, which the rater treats as ties rather than breaking arbitrarily.

## 2 The rater

Each driver carries a belief over latent ability: a whole density on a uniform lattice, free to be skewed or multimodal, diffusing between races in proportion to elapsed time. A race performance is ability plus noise drawn from a *base density* — any density; this freedom is the paper’s subject. Predictions price the joint order statistics of the field exactly (dead-heat aware,  $O(N)$  per race) using the machinery of the fast ability transform. Updates use the exact likelihood of the observed finish order in each driver’s ability, with opponents at their predictive marginals: with finishers indexed fastest first, a forward chain accumulates prefix sums of (density  $\times$  chain) over the faster finishers, a backward chain does the same over the slower ones, and the  $k$ -th finisher’s likelihood is the base density correlated with the product of the two chains. One coherent expression per driver,  $O(N)$  overall, validated against brute-force simulation. An earlier version of the system factorized the order into peeled stages and multiplied overlapping factors; the chain replaced it after a paired test showed the factorization was measurably lossy — worse everywhere, at twice the cost.

## 3 The benchmark

All systems observe identical data prequentially: the first 20% of races are unscored warm-up; every later race is predicted before it is observed. Log loss, Brier and accuracy score the winner prediction; Kendall’s tau the predicted order; expected calibration error (ECE) the pooled win probabilities; and a randomized rank-PIT statistic measures whether each system’s *distribution* over finish positions matches reality. TrueSkill receives the empirical dead-heat rate; comparison rows use each package’s own prediction formula.

| system                        | log loss     | accuracy    | tau         | ECE  | rank-PIT |
|-------------------------------|--------------|-------------|-------------|------|----------|
| Thurstone lattice             | <b>2.225</b> | <b>.316</b> | <b>.328</b> | .015 | .176     |
| TrueSkill                     | 2.346        | .275        | .300        | .011 | .199     |
| Elo (multi-entrant)           | 2.465        | .303        | .319        | .026 | .220     |
| OpenSkill Plackett–Luce       | 2.705        | .259        | .283        | .023 | .223     |
| OpenSkill Thurstone–Mosteller | 2.951        | .007        | .106        | .010 | .628     |
| uniform baseline              | 3.162        | .043        | —           | —    | —        |
| Glicko-2                      | 3.727        | .271        | .255        | .022 | .166     |

Two failure modes deserve notice. Glicko-2, sound at head-to-head scale, scores *below the uniform baseline*: flooding a single rating period with nineteen correlated pairwise results drives its volatility machinery to numerical extremes (our implementation already includes stabilizing guards; without them it does not finish the century). OpenSkill’s Thurstone–Mosteller variant, which decomposes the order into all pairs, diverges similarly. The lesson is structural: pairwise decomposition is not a harmless approximation at grid scale.

## 4 The performance density is a free parameter

Every comparator above is locked to one family — Gaussian beliefs with Gaussian or Gumbel noise. The lattice’s base density is an array, so the noise model becomes an empirical question. We swept it.

| base density                     | log loss     | ECE         | rank-PIT    |
|----------------------------------|--------------|-------------|-------------|
| $N(0, 1) + 25\%$ disaster slab   | 2.203        | .015        | .158        |
| era-adaptive slab                | <b>2.200</b> | .016        | .185        |
| Gaussian                         | 2.225        | .015        | .176        |
| skew-normal ( $a=2$ )            | 2.222        | <b>.010</b> | .165        |
| Student- $t$ ( $\nu=5$ )         | 2.271        | .018        | .171        |
| tight core (0.5) + 25% slab      | 2.254        | .013        | <b>.136</b> |
| near-Dirac core (0.2–0.3) + slab | 2.42–2.52    | —           | —           |

The *disaster slab* — ordinary Gaussian pace with probability  $1 - p$ , a uniform lump of far-slow mass with probability  $p$  — matches the physics of retirement: a blown engine is not a slow day, it is a separated event that no level of skill outruns. It improves every metric.

Three hypotheses failed, and the pattern of failure is the finding. Symmetric heavy tails ( $t$ -family) hurt monotonically as tails fatten: symmetry forces miracle mass onto the *fast* side, and nobody outruns their ability by five sigma. Skew-normal noise does approximately nothing here, as it did on every dataset we tried; finish orders are blind to smooth asymmetry (pairwise comparisons cancel skew exactly, and ranks are invariant to monotone transformations of the clock). And a near-Dirac pace core — the hypothesis that finishing order among finishers is nearly deterministic — is refuted at 0.2 log loss: strategy, traffic and weather are order-one noise. A moderately tight core does, however, produce the best distributional calibration we measured (rank-PIT .136), so winner prediction and field-shape prediction pull the kernel in different directions: a genuine Pareto frontier, of which the table shows three corners.

## 5 The slab is physics

Fixing  $p$  at 25% is itself a Gaussian-era compromise. The era-adaptive variant estimates  $p$  prequentially from a trailing window of observed retirement frequency (no leakage: past races only) and rebuilds the base density on the fly. It sets the overall record (log loss 2.2001, best Kendall’s tau), and its fitted trajectory is the point: the mass falls across seven decades and ends near the modern retirement rate. The parameter is not a tuning constant; it is a measurement of the sport, made by the rating system in passing.

## 6 What we did not do

Driver-versus-car identification is unaddressed: constructor effects are folded into driver ratings, and separating them (each driver a latent vector, the car a direction) is the natural next study. Retirements enter tied at the back, discarding the official ordering of retirees by laps completed; the alternative is an untested A/B. We did not test against a betting market, for want of public historical odds with full-field coverage. And the ratings themselves are one-dimensional: no wet-weather axis, no circuit-type axis, though the machinery for both exists.

## Reproducibility

Everything is one command per table from the `winning` package ([github.com/microprediction/winning](https://github.com/microprediction/winning)):  
`python -m winning.benchmarks.run_benchmark --dataset f1` for the benchmark, and `research/f1_dirac_`  
`research/t_sweep.py`, `research/f1_era_slab.py` for the density studies. Data is fetched at  
run time from `f1db` (CC-BY-4.0). The underlying lattice machinery is the `thurstone` package;  
the ability transform is described in Cotton, *SIAM J. Financial Mathematics* 12(1), 2021.